METRIC
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SUPERSEDING
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DETAIL SPECIFICATION

BUTYL SEBACATE, DI-NORMAL

This specification is approved for use by the Naval Surface Warfare Center Indian Head Division (NSWC IHD) and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for one type of di-normal butyl sebacate (di-n-butyl sebacate) suitable for use in the manufacture of propellants.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

COMMERCIAL ITEM DESCRIPTIONS

A-A-59282 - Chemicals, Analytical; General Specification for

(Copies of this document are available online at https://quicksearch.dla.mil.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

NAVAL SEA SYSTEMS COMMAND (NAVSEA) PUBLICATIONS

T9070-AL-DPC-020/077-2 - NAVSEA Hazardous Material Avoidance Process

(Copies of this document are available online via Technical Data Management Information System (TDMIS) at https://mercury.tdmis.navy.mil by searching for the document number without the suffix. Refer questions, inquiries, or problems to: DSN 296-0669, Commercial (805) 228-0669. This document is available for ordering (hard copy) via the Naval Logistics Library (NLL) at https://nll.navsup.navy.mil. For questions regarding the NLL, contact the NLL Customer Service at nllhelpdesk@navy.mil, (866) 817-3130, or (215) 697-2626/DSN 442-2626.)

Comments, suggestions, or questions on this document should be addressed to Commander, Naval Sea Systems Command, ATTN: SEA 05S, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard, DC 20376-5160 or emailed to CommandStandards@navy.mil, with the subject line "Document Comment". Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

AMSC N/A FSC 6810

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NAVAL SURFACE WARFARE CENTER INDIAN HEAD DIVISION (NSWC IHD) PUBLICATIONS

Lab Procedure 310-301 - Determination of Sodium Content in Di-n-butyl Sebacate (DBS) by the Atomic Absorption Method

Lab Procedure AAL-19001 - Analysis for Purity of DBS by Capillary Gas Chromatography with Flame Ionization Detection (GC-FID)

(Copies of these documents are available from Naval Surface Warfare Center Indian Head Division, Code E16, 3703 Benson Road, Indian Head, MD 20640-5116.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AEROSPACE INDUSTRIES ASSOCIATION (AIA)

NAS411-1 - Hazardous Materials Target List

(Copies of this document are available online at www.aia-aerospace.org.)

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI/ASQ Z1.4 - Sampling Procedures and Tables for Inspection by Attributes

(Copies of this document are available online at https://webstore.ansi.org.)

ASTM INTERNATIONAL

ASTM D1209 - Standard Test Method for Color of Clear Liquids (Platinum-Cobalt Scale)

ASTM D1218 - Standard Test Method for Refractive Index and Refractive Dispersion of Hydrocarbon Liquids

ASTM D4057 - Standard Practice for Manual Sampling of Petroleum and Petroleum Products

ASTM E29 - Standard Practice for Using Significant Digits in Test Data to Determine Conformance with Specifications

ASTM E203 - Standard Test Method for Water Using Volumetric Karl Fischer Titration

(Copies of these documents are available online at www.astm.org.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Chemical and physical properties. The chemical and physical properties of the di-n-butyl sebacate shall conform to those listed in table I when tested as specified herein (see 4.2).

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TABLE I. Chemical and physical properties.

Requirements			
Property	**Minimum**	**Maximum**	**Test Method**
Color (Pt-Co scale)	-	10	4.2.1
Acidity (as acetic acid) (% by wt)	-	0.05	4.2.2
Saponification number	350	360	4.2.3
Refractive index (n_D25 °C)	1.4385	1.4405	4.2.4
Moisture (% by wt)	-	0.15	4.2.5
Sodium (ppm)	-	1.0	4.2.6
Purity (% by wt)	-	98.5	4.2.7

3.2 Workmanship. The di-n-butyl sebacate shall be a colorless, homogeneous liquid, free from dirt, sediment, and other suspended foreign matter when examined visually by transmitted light (see 4.2.8).

3.3 Limiting values. The following applies to all specified limits in this specification. For purposes of determining conformance with these requirements, an observed value or a calculated value shall be rounded off to the nearest unit in the last right-hand digit used in expressing the specification limit according to the rounding method in accordance with ASTM E29.

3.4 Prohibited materials. The di-n-butyl sebacate shall not contain any chemicals categorized as “prohibited” in accordance with T9070-AL-DPC-020/077-2 for NAVSEA programs or NAS411-1 for all other programs.

4. VERIFICATION

4.1 Conformance inspection. The material shall be subjected to all the inspections and tests for acceptance listed in 4.2. When specified (see 6.2), the supplier shall submit a report giving the results obtained for all inspections and tests performed and a certified statement that the lot meets all the requirements of this specification. Unless otherwise specified (see 6.2), all chemicals shall be American Chemical Society (ACS) grade in accordance with A-A-59282, and distilled water shall be used. Where applicable, blank determinations shall be run and corrections applied where significant. If the individual or composite specimens (see 4.1.3) fail to conform to any of the requirements specified, the lot shall be rejected.

4.1.1 Lot formation. Lot formation shall be performed in accordance with the lot formation requirements of ANSI/ASQ Z1.4, paragraph “Formation and identification of lots or batches.” An inspection lot refers only to an initial production lot or any combined lot, offered for acceptance at one time, that meets all requirements specified in 4.2 from the vendor and does not apply to any lots combined in storage or shipping by the plant. If the process is a batch operation (see 6.3), then each batch shall be a lot. If the di-n-butyl sebacate is produced in a continuous process, a lot shall consist of material produced by one manufacturer, at one plant, with no change in the process, materials, or conditions, and under essentially the same manufacturing conditions or an alternate manufacturing plan approved by the procuring activity.

4.1.2 Sampling. Sampling from tank cars shall be conducted in accordance with ASTM D4057. Sampling from smaller containers shall be conducted as follows: A random sample shall be taken from each lot. Sample size shall be determined in accordance with ANSI/ASQ Z1.4, inspection level S2. There shall be no defects in the sample. Any defects shall be cause for rejection of the lot. If there are fewer than three containers in a lot, each container shall be sampled. In all other cases, no fewer than three containers shall be selected. A 500-milliliter specimen shall be removed from each container in the sample and placed in a clean, dry container (nonreactive with di-n-butyl sebacate) with a tight-fitting closure (e.g., a 1-quart mason jar). The container shall be labeled to identify the lot and container from which it was taken. Sampling shall be conducted in such a manner that atmospheric exposure of the contents of the shipping container and the sample is minimized.

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4.1.3 Test specimens. A composite specimen shall be made with equal portions from each specimen and the composite shall be tested as specified in 4.2. If there are fewer than three specimens, each one shall be tested as specified in 4.2.

4.2 Methods of inspection.

4.2.1 Color. The color of di-n-butyl sebacate shall be determined in accordance with ASTM D1209, platinum cobalt scale.

4.2.2 Acidity. The acidity shall be determined as follows:

a. Measure 100 milliliters of ethyl alcohol into a flask, add three drops of phenolphthalein indicator, and titrate the alcohol to a faint pink color with 0.1N sodium hydroxide solution.

b. Add an accurately weighed 30- to 40-gram portion of the sample to the alcoholic solutions.

c. Titrate the mixture with 0.1N sodium hydroxide

d. Calculate the percentage of acidity as acetic acid as follows.

$$Percent\ by\ wt\ acetic\ acid = \frac{6.005\ VN}{G}$$

Where:

V = sodium hydroxide required for titration of sample, milliliters

N = normality of sodium hydroxide solution

G = weight of sample, grams

4.2.3 Saponification number. The saponification number shall be determined as follows:

a. Transfer 0.8 gram weighed to ±0.1 milligram of the sample to an appropriate-sized flask, and add 50 milliliters of 0.5N alcoholic potassium hydroxide by means of a calibrated burette or pipette. (0.5N alcoholic potassium hydroxide is prepared by adding 28 grams of potassium hydroxide to 1 liter of 95 percent ethyl alcohol.)

b. Fit the flask to a reflux condenser by means of ground joints and heat on a water bath until the precipitation appears to be complete.

c. Add approximately 15 milliliters of water to dissolve the potassium sebacate and reflux on a water bath for not less than 60 minutes with occasional whirling of the flask.

d. Wash down the sides of the reflux condenser and the ground joints with approximately 25 milliliters of the water.

e. Cool the solution to 25 °C and titrate with standard 0.5N hydrochloric acid using phenolphthalein indicator.

f. Protect the sample solution from carbon dioxide during heating and cooling with an Ascarite, or equivalent, guard tube.

g. At the same time, conduct a blank determination on 50 milliliters of alcoholic potassium hydroxide that has been carried through the complete process.

h. Calculate the saponification number as follows:

$$Saponification\ number\ (milligrams\ potassium\ hydroxide\ per\ gram) = \frac{(V_1 - V_2)\ x\ N\ x\ 56.1}{W}$$

Where:

V_1 = volume of hydrochloric acid solution required for blank titration, milliliters

V_2 = volume of hydrochloric acid solution required for the sample titration, milliliters

N = normality of the hydrochloric acid solution

W = weight of sample, grams

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4.2.4 Refractive index. The refractive index shall be determined in accordance with ASTM D1218.

4.2.5 Moisture. The moisture content shall be determined in accordance with ASTM E203.

4.2.6 Sodium. The sodium content shall be determined in accordance with NSWC IHD Lab Procedure 310-301, noting that a factor of 1.33 shall be used. An alternate procedure approved by the procuring activity prior to use may be used (see 6.2).

4.2.7 Purity. The purity shall be determined in accordance with NSWC IHD Lab Procedure AAL-19001.

4.2.8 Visual inspection. Visual inspection of the product to ensure the material conforms to the workmanship requirement of 3.2 shall be performed after a representative aliquot of the sample has been transferred to a clear glass vessel and compared to that of the same type of vessel containing the same volume of purified water.

4.3 Prohibited materials. Prohibited materials shall be verified as specified (see 3.4 and 6.2).

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The di-n-butyl sebacate covered by this specification is intended for use in the manufacture of propellant as it applies to torpedo propulsion. Successful utilization of plasticizers in propellant and explosive formulations is dependent on the specific properties of the plasticizer, which constitute the requirements in this specification. Current stock of di-n-butyl sebacate may be used until depleted; future requisitions for di-n-butyl sebacate should refer to the current version of this specification.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Requirement for prohibited materials verification (see 3.4 and 4.3).
c. Whether a certified analysis and/or test report is required (see 4.1).
d. Grade of chemicals if other than ACS (see 4.1).
e. Alternate test methods for determining sodium content, when appropriate (see 4.2.6).
f. Packaging and packing requirements (see 5.1 and 6.5).
g. Activities that require safety data sheets (see 6.4).
h. Size of containers required (see 6.5).
i. Whether tank cars or containers are to be used for shipment (see 6.5.1).
j. Sections of 49 CFR specifying container requirements (see 6.5.1).
k. Marking requirements (see 6.6).

6.3 Batch. A batch is defined as the quantity of di-n-butyl sebacate that has been subjected to the same unit chemical or physical process intended to make the final product homogeneous (see 4.1.1).

6.4 Safety data sheets. Contracting officers will identify those activities that require copies of completed safety data sheets prepared in accordance with 29 CFR 1910.1200 (Hazard Communication) (see 6.2).

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6.5 Packaging and packing. Unless otherwise specified (see 6.2), packaging and packing should be level C.

6.5.1 Level C. The di-n-butyl sebacate should be packaged in uniform quantities for shipment per the manufacturer's commercial practice. Packages should be of uniform size, shape, and materials. The di-n-butyl sebacate should be packed in uniform quantities in such a manner as to assure carrier acceptance and afford protection against damage during direct shipment from the supply source to the first receiving activity. Unless otherwise specified (see 6.2), packaging and packing of the plasticizer should conform with method 10 of MIL-STD-2073-1. The packaging and packing should be of such construction and materials that the contents will be adequately protected against loss or contamination. Container size should be as specified (see 6.2). Containers should conform to the sections of 49 CFR, as specified (see 6.2).

6.6 Marking. In addition to any special marking required by the contract (see 6.2), unit packages and shipping containers should be marked in accordance with MIL-STD-2073-1.

6.7 Subject term (key word) listing.

DBS

Desensitizer

Monopropellant

Propellant

Stabilizer

6.8 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

CONCLUDING MATERIAL

Custodian:
Navy – OS

Review activity:
DLA – GS

Preparing activity:
Navy – OS
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